

IT3051 – Fundamentals of Data Mining

Mini Project 2026

Dataset Proposal Document

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Hotel Booking Cancellation Prediction (Hospitality / Revenue Management)

1. Dataset Name and Original Source

Dataset name	Hotel Booking Demand (Hotel Booking Demand Datasets)
Original source	Antonio, de Almeida & Nunes. Real booking records from two hotels (a resort hotel and a city hotel) for bookings due to arrive between 1 July 2015 and 31 August 2017, including bookings that were cancelled. Hotel and customer identification data were removed by the authors. Published as a data article in Data in Brief (2019).
Platform used	Kaggle (uploaded by jesse mostipak), file hotel_bookings.csv. The original source is the data article cited in Section 2.

2. Dataset URL / Download Source and Citation

Kaggle page	https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand
Original article (DOI)	https://doi.org/10.1016/j.dib.2018.11.126
Direct download	https://www.kaggle.com/api/v1/datasets/download/jessemostipak/hotel-booking-demand (Kaggle account required). Kaggle CLI: kaggle datasets download -d jessemostipak/hotel-booking-demand
Citation	Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126
File	hotel_bookings.csv (119,390 rows × 32 columns).
License	Released with an open-access data article and redistributed on Kaggle. Creative Commons Attribution 4.0 International (CC BY 4.0), as listed on the Kaggle dataset page. https://creativecommons.org/licenses/by/4.0/

3. Problem Scenario, Dataset Description and Real-World Context

3.1 Real-world problem

Hotels lose revenue when bookings are cancelled: rooms stay empty or have to be resold late at lower prices. To protect revenue, hotels overbook, but overbooking too much means guests may be turned away, which is costly. A revenue or reservations manager therefore needs to know, at the time a booking is made, how likely it is to be cancelled.

Scenario: a booking cancellation early-warning system for a hotel. Given the details of a new booking (hotel, lead time, arrival date, length of stay, guests, market segment, deposit type, customer type, booking history and so on), the system predicts whether the booking will be cancelled and reports the probability of cancellation.

Prediction time: the moment the booking is made. Only information available at that moment may be used as a predictor.

3.2 Stakeholders and decisions supported

- **Revenue manager:** decides how many rooms to overbook per night and how to price.
- **Reservations team:** decide which risky bookings to contact for confirmation or to require a deposit.
- **Hotel management:** understands what drives cancellations (market segment, deposit type, lead time, customer type) to improve booking policies.

3.3 Dataset description

Each row is one hotel booking. The data covers two hotels, a resort hotel (40,060 bookings) and a city hotel (79,330 bookings), with arrivals from July 2015 to August 2017. Each booking records when and how it was made, the stay dates and length, the guests, meal plan, guest country, market segment and distribution channel, deposit type, customer type, booking history, room details, price (average daily rate), special requests, and whether the booking was cancelled.

4. Number of Records and Relevant Features

Records: 119,390 bookings.

Columns: 32 (the target is `_canceled` and 31 other columns).

Candidate predictors: about 28 after removing the leakage columns.

Feature(s)	Type	Description
hotel	Categorical	City Hotel or Resort Hotel.
lead_time	Numeric	Days between booking date and arrival date.
arrival_date_year, _month, _week_number, _day_of_month	Mixed	Components of the arrival date (month is stored as a name).
stays_in_weekend_nights, stays_in_week_nights	Numeric	Number of weekend and weekday nights booked.
adults, children, babies	Numeric	Number of guests of each type.
meal	Categorical	Meal plan (BB, HB, FB, SC, Undefined).
country	Categorical	Guest country code (177 distinct values).
market_segment, distribution_channel	Categorical	How and through whom the booking was made.
is_repeated_guest, previous_cancellations, previous_bookings_not_canceled	Binary / numeric	Guest booking history.
reserved_room_type, assigned_room_type	Categorical	Room type requested and room type assigned (coded).
booking_changes	Numeric	Number of changes made to the booking.
deposit_type	Categorical	No Deposit, Non Refund or Refundable.
agent, company	ID (categorical)	Travel agency and company that made the booking (many missing).

Feature(s)	Type	Description
days_in_waiting_list	Numeric	Days the booking was on a waiting list before confirmation.
customer_type	Categorical	Contract, Group, Transient or Transient-Party.
adr	Numeric	Average daily rate.
required_car_parking_spaces, total_of_special_requests	Numeric	Parking spaces and special requests.
reservation_status, reservation_status_date	Categorical / date	Final status of the booking and the date it was set. Leakage: excluded (Section 8).
is_canceled	Binary (target)	1 = cancelled or no-show, 0 = stayed.

Planned feature engineering: total nights, total guests, family flag, reserved-vs-assigned room match (if retained), arrival season and weekday, lead-time bands, previous-cancellation rate, and grouped country and agent categories.

5. Target Variable and Its Meaning

Target: is_canceled. **Type:** binary (1 = the booking was cancelled or the guest did not show up, 0 = the guest checked out).

In the raw file, 44,224 bookings (37.0%) have the value 1 (43,017 cancellations and 1,207 no-shows) and 75,166 (63.0%) have the value 0. This is moderate class imbalance. The rate is higher at the city hotel (about 41.7%) than at the resort hotel (about 27.8%).

6. Expected Machine Learning Task

Task type: supervised binary classification, with the predicted probability of cancellation reported to the user.

Planned algorithms

- Logistic Regression: interpretable baseline.
- Decision Tree and Random Forest: capture non-linear effects and interactions between segment, deposit type and lead time.
- Gradient Boosting (XGBoost or LightGBM): expected to perform strongly on tabular data.
- K-Nearest Neighbours and SVM: a further model family for comparison, run on a stratified sample if training time requires it.

Validation and metrics

- A stratified train/test split, plus a time-based check (train on earlier arrival dates and test on later ones) to see how the model behaves on future bookings.
- Stratified k-fold cross-validation with all preprocessing fitted on training folds only.
- Accuracy alone is not enough for an imbalanced target. The team will report recall, precision and F1 for the cancellation class, ROC-AUC, PR-AUC and confusion matrices, and will choose the decision threshold using the relative cost of an empty room (missed cancellation) versus overbooking (false alarm).

- Hyperparameter tuning with randomized or grid search (on a stratified sample where needed for speed), comparing tuned models with their baselines.

7. Reason Why the Dataset Is Suitable

The dataset records exactly what the scenario needs: the details a hotel holds when a booking is made, and the eventual outcome. It was chosen because it fits the scenario and offers real data-mining depth, not because it is easy to use. The table below maps it to the dataset selection guidelines in the assignment brief.

Guideline	How the dataset meets it
Relevant to the scenario	Booking details at reservation time and the cancellation outcome are exactly what a revenue manager needs to model.
Clear target variable	Binary target: <code>is_canceled</code> .
Sufficient observations and features	119,390 bookings and about 30 features, enough for stable cross-validation, tuning and a time-based check.
Credible, public source	Real hotel data published with a peer-reviewed data article (Data in Brief, 2019).
Accessible to all members	Single CSV file that every member can download for free.
Enough complexity for EDA, preprocessing, modelling	Real missing values, hidden missing codes, duplicates, invalid records, outliers, high-cardinality categorical, imbalance and a clear leakage risk.
Limited privacy, legal or ethical risk	Hotel and customer identifiers were removed by the authors. Sensitive-feature handling is described in Section 9.

8. Important Data-Quality Observations and Known Limitations

The following was observed in the raw file during the team's initial check and will be examined in detail in EDA.

Data-quality issues and planned handling

Issue	Observation in the raw file	Planned handling
Missing values	company missing in 112,593 rows (about 94%), agent in 16,340 (about 14%), country in 488 (about 0.4%), children in 4.	Drop company or convert it to a "has company" flag. Treat a missing agent as "no agent". Impute country with a category or the mode. Fill children with 0.
Hidden missing values	meal has 1,169 "Undefined" values, market_segment 2 and distribution_channel 5.	Recode as a known category (e.g. meal "Undefined" is equivalent to "SC", no meal package) or group into "other", and document the reasoning.
Duplicate records	31,994 rows (about 27%) are exact duplicates across all columns.	Investigate whether these are repeat bookings or true duplicates, then decide and justify. Ensure identical

Issue	Observation in the raw file	Planned handling
		rows are not split across train and test.
Invalid records	180 bookings with zero guests, 715 with zero nights, 1 negative adr and adr up to 5,400.	Remove or correct clearly invalid rows and document how many were affected.
Outliers	lead_time up to 737 days, adults up to 55, babies up to 10, extreme adr.	Inspect boxplots and the IQR rule. Cap or transform for scale-sensitive models and leave trees unchanged.
High cardinality	country has 177 values, agent 333 IDs and company 352 IDs.	Group rare values or use frequency or target encoding fitted on training data only.
Class imbalance	37.0% cancelled vs 63.0% not cancelled.	Compare class weights and resampling, applied inside training folds only.

Unusual pattern to investigate : bookings with deposit_type “Non-Refund” (14,587 rows) are cancelled 99.4% of the time, which is counter intuitive. It may reflect a real business pattern or a data artefact. The team will investigate it and compare model performance with and without deposit_type.

Potential data leakage and prevention

- **reservation_status and reservation_status_date:** reservation_status determines the target exactly (every “Canceled” and “No-Show” is 1, every “Check-Out” is 0). Both columns will be excluded.
- **assigned_room_type:** bookings where the assigned room differs from the reserved room are cancelled only 5.4% of the time, against 41.6% otherwise. This suggests the room is assigned close to arrival, after the cancellation outcome is largely known, so it will be excluded unless shown to be available at booking time.
- **booking_changes and days_in_waiting_list:** these may be recorded after the booking is made (bookings with changes are cancelled 15.7% of the time versus 40.9% without). Each will be reviewed against the stated prediction time and kept or excluded with justification.
- The train/test split will be done before any fitting. Imputation, encoding, scaling, feature selection and resampling will be fitted on training data only, inside cross-validation folds.

Other limitations

- The data covers two hotels between 2015 and 2017, so results may not generalize to other hotels or periods. The project demonstrates a method that could be retrained on a hotel’s own current data.
- The dataset has no cost or revenue figures, so the cost-based threshold analysis will use clearly stated assumptions.

9. Ethical, Privacy, Licensing and Accessibility Considerations

- **Privacy:** the authors removed all hotel and customer identification data. The dataset contains no names or contact details.
- **Fairness:** country is a proxy for nationality. The team will compare model performance with and without it and examine error rates across country groups. The system will be framed as decision support for operational planning (overbooking, confirmation outreach and deposits) and not for refusing bookings based on nationality.
- **Licensing:** the dataset will be credited using the citation in Section 2 in the report, code and presentation, and the license will be stated as shown on the source page.
- **Accessibility:** a single CSV file freely downloadable from Kaggle (a free account is needed), and the source data article is open access.

10. Proposed System Overview

- **Backend (Python, FastAPI):** loads the final trained model and preprocessing pipeline, validates inputs (required fields, allowed categories, numeric ranges, sensible dates), applies the same feature engineering used in training, and returns the cancellation probability and risk band.
- **Frontend (React):** a booking form with clear labels, dropdowns for categorical fields, inline validation and helper text. A batch CSV upload for a list of bookings is a possible extension.

Expected inputs: hotel, booking lead time or booking and arrival dates, number of weekend and weekday nights, adults, children, babies, meal plan, guest country, market segment, distribution channel, deposit type, customer type, repeated-guest flag, previous cancellations, requested room type, expected daily rate, parking spaces and special requests.

Expected output: the probability that the booking will be cancelled, a risk band (low, medium or high, based on a threshold chosen from the cost analysis), and a suggested action such as “request a deposit” or “send a confirmation reminder”, with the main factors behind the prediction.